REVIEW 2

AI-DRIVEN SMART WORKER SAFETY AND EMERGENCY COORDINATION SYSTEMFOR PREDICTIVE HEALTH MONITORING AND INTELLIGENT RESOURCE MANAGEMENT

TEAM MEMBERS :

ABINISHASRI B - 927623BAD002

DHARSHINI K - 927623BAD023

GOPIKASRI T - 927623BAD037

PROJECT GUIDE:

Dr. K .V.NITHYA

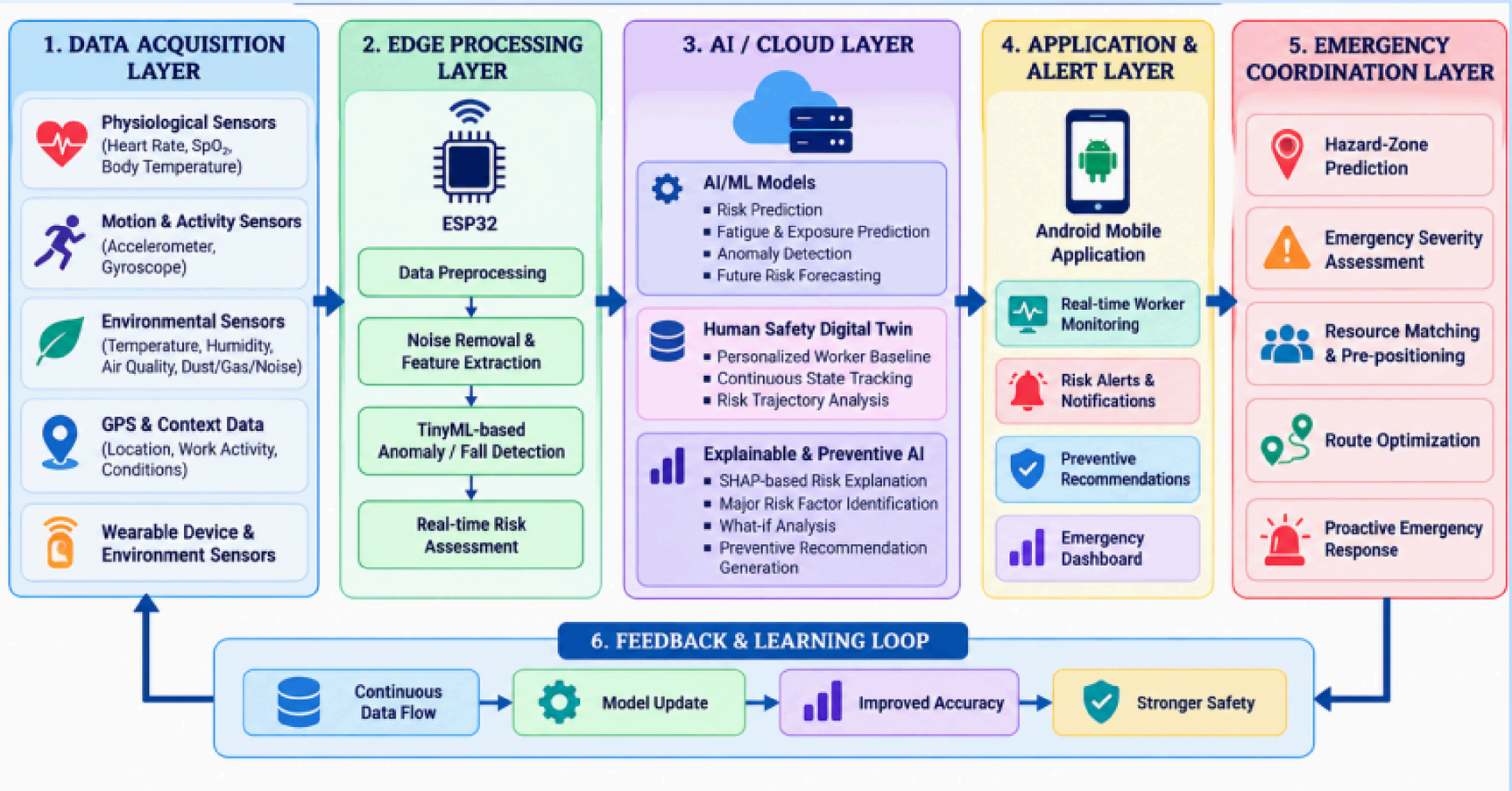
TEAM NO. : 4

DATE : 12.10.2026

PROBLEM OVERVIEW

- An AI-driven worker safety system for continuous and personalized safety monitoring.
- Collects physiological, activity, environmental, and contextual data from workers.
- Uses AI/ML models to predict current and future safety risks, fatigue, exposure, and abnormal conditions.
- Creates a Human Safety Digital Twin for personalized risk assessment.
- Uses Explainable AI to identify major risk factors and provide preventive recommendations.
- Supports proactive emergency response through hazard-zone prediction, severity assessment, resource coordination, and route optimization.

FINALIZED SYSTEM DESIGN



IMPLEMENTATION APPROACH

- Data Collection: ESP32 with physiological, motion and environmental sensors.
- Edge Processing: Data preprocessing, noise removal and feature extraction.
- AI/ML: Risk prediction, fatigue/exposure prediction and anomaly detection.
- Digital Twin: Personalized worker safety profile and continuous risk tracking.
- Explainable AI: SHAP-based risk factor identification and What-if analysis.
- Application & Cloud: Android application, Firebase and FastAPI for real-time monitoring and alerts.
- Emergency Response: Hazard-zone prediction, severity assessment, resource matching and route optimization.

IMPLEMENTATION MODULES/COMPONENTS

1.Multimodal Data Acquisition:

- Physiological sensors
- Motion & activity sensors
- Environmental sensors
- GPS & context data

2.Edge Processing & Digital Twin:

- ESP32-based preprocessing
- Noise removal & feature extraction
- TinyML anomaly/fall detection
- Personalized worker safety profile

IMPLEMENTATION MODULES/COMPONENTS

3.AI/ML Risk Prediction:

- Current risk prediction
- Fatigue & exposure prediction
- Anomaly detection
- Future risk forecasting

4.Explainable & Preventive AI:

- SHAP-based risk explanation
- Risk factor identification
- What-if analysis
- Preventive recommendations

IMPLEMENTATION MODULES/COMPONENTS

5.Emergency Coordination:

- Hazard-zone prediction
- Emergency severity assessment
- Resource matching & pre-positioning
- Route optimization

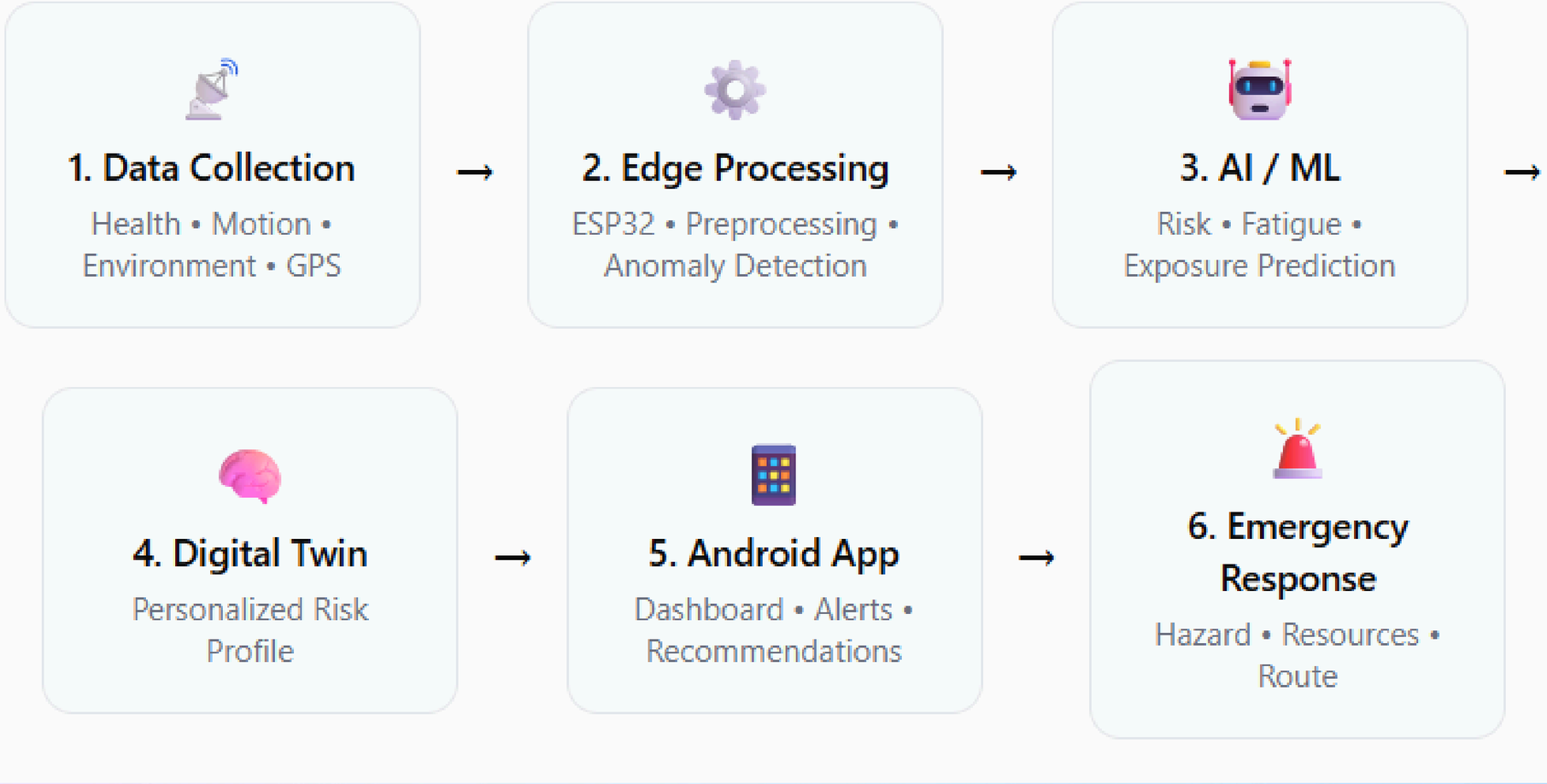
6.Android & Cloud Integration:

- Android mobile application
- Real-time monitoring
- Alerts & notifications
- Emergency dashboard

CORE IMPLEMENTATION /WORKING PROTOTYPE

- ESP32-based sensor integration for collecting worker health, activity and environmental data.
- Real-time data preprocessing including noise removal and feature extraction.
- TinyML-based anomaly/fall detection at the edge.
- AI/ML risk prediction for identifying current and future worker safety risks.
- Human Safety Digital Twin for personalized worker monitoring and risk tracking.
- Android dashboard for displaying real-time worker status, risk alerts and recommendations.
- Emergency coordination for hazard prediction, resource matching and proactive response.

SYSTEM INTEGRATION/WORKING FLOW



INITIAL RESULTS/OUTPUT

- Sensor data collection from physiological, motion and environmental sensors.
- Preprocessed sensor data ready for AI/ML analysis.
- Real-time worker safety parameters can be monitored.
- Risk prediction output for identifying potential safety conditions.
- Personalized worker safety profile through the Digital Twin.
- Risk factors and preventive recommendations using Explainable AI.
- Real-time safety alerts and dashboard output through the application.
- Emergency severity and resource coordination output for critical situations.

TESTING & VALIDATION

- Sensor Testing – Verify accurate collection of physiological and environmental data.
- Data Validation – Check data quality, preprocessing and feature extraction.
- ML Model Testing – Evaluate risk prediction and anomaly detection.
- Digital Twin Validation – Verify personalized worker safety profile updates.
- Application Testing – Test dashboard, alerts and recommendations.
- Emergency Module Testing – Validate hazard prediction, resource matching and response flow.
- System Integration Testing – Verify complete flow from sensors → AI/ML → application → emergency response.

CURRENT PROGRESS & LIMITATIONS

Current Progress:

- System architecture finalized.
- Six major modules finalized.
- Hardware components selected.
- ML algorithms and technologies selected.
- Initial system design completed.
- Firebase database and Android application structure planned.

Limitations:

- Real-world worker sensor data is not yet fully available.
- Complete hardware integration is still pending.
- ML model evaluation requires further testing.
- Android and cloud integration needs to be completed.
- Full emergency coordination and system validation are yet to be completed.

PHASE 1 COMPLETION &PHASE 2 PLAN:

Phase-I Completion:

- Project problem and research gap finalized.
- System architecture and six modules finalized.
- Hardware and software components selected.
- AI/ML implementation approach finalized.
- Android Mobile Application completed.
- Mobile app interface and major functionalities implemented.
- Firebase/Cloud integration structure prepared.

PHASE 1 COMPLETION &PHASE 2 PLAN:

Phase-II Plan:

- Complete hardware and sensor integration.
- Connect real-time sensor data with the mobile application.
- Implement and evaluate AI/ML risk prediction models.
- Integrate Digital Twin and Explainable AI.
- Implement emergency coordination and resource optimization.
- Perform complete system testing and validation.
- Evaluate the final integrated prototype.

THANK YOU